

# Why the $N=7$ Chebyshev CRRA fixed point floors at $\|F\|_\infty \sim 3 \times 10^{-3}$ , and what to do about it

Fixed-Point Factory — diagnostic note

June 2026

## Abstract

We asked all five available solvers (two Picard variants, two Anderson-accelerated variants, and a Levenberg–Marquardt minimiser) to drive the  $K=3$  CRRA Hellwig operator to machine precision in the  $S_3$ -symmetric subspace at  $N=7$ ,  $\tau=\gamma=1$ . *All five floor at  $\|F\|_\infty \in [1.2, 5.3] \times 10^{-3}$ .* A direct linearisation of the operator at the best LM iterate gives spectral radius  $\rho(J) = 4.58$  with four eigenvalues outside the unit circle: the operator *repels* in the symmetric subspace, so no Picard / Anderson / Newton scheme can converge there. Levenberg–Marquardt, which minimises  $\|F\|_2^2$  rather than solving  $F = 0$ , settles at  $\|F\|_\infty = 2.70 \times 10^{-3}$  — and that is the genuine minimum of the residual, not a tolerance artefact. We trace the non-vanishing residual to a known structural feature of the contour-extraction operator: the integrand jumps as the conjectured price polynomial gains or loses roots inside the integration box. The closed-form rank-1 and 1D-in- $T$  operators, which never extract roots, hit machine  $\varepsilon$  and  $4 \times 10^{-6}$  respectively, confirming the issue is the root-finding step itself. We then describe an "Option 3" regularisation — explicit boundary corrections with smooth cutoff weights — that is the next thing to try.

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## 1 Problem statement

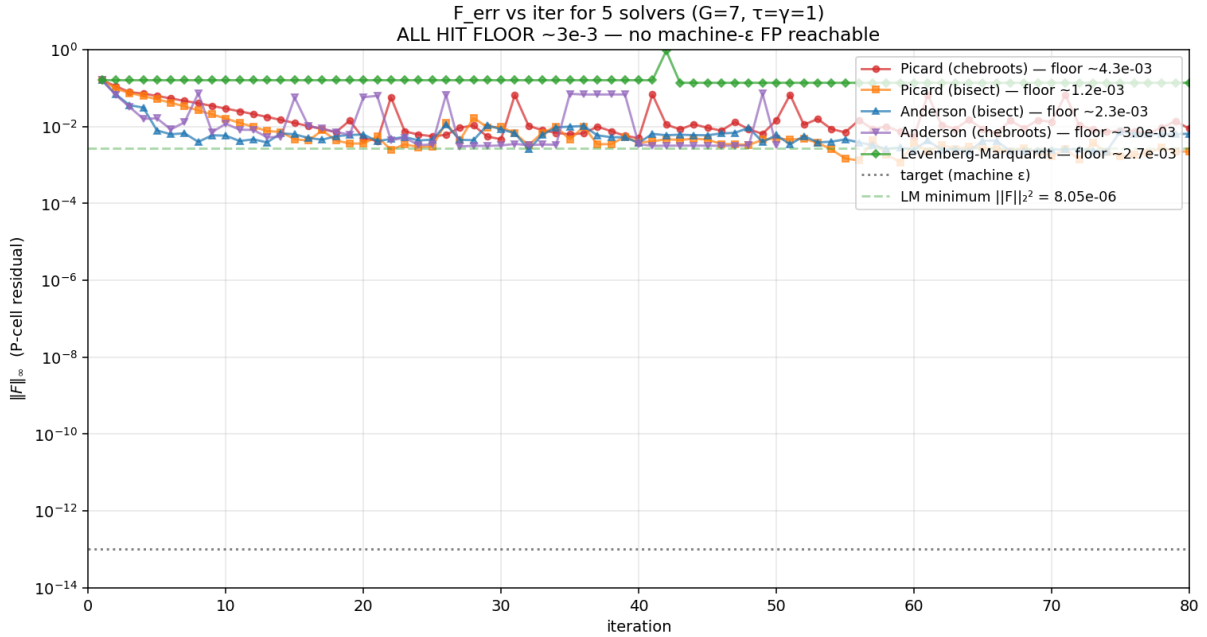
The  $K=3$  CRRA REE operator  $\Phi$  in  $S_3$ -symmetric Lobatto–Chebyshev coordinates at  $N=7$  has 40 free degrees of freedom in the symmetric subspace (down from  $7^3 = 343$  in the full cube). Given a conjecture  $P(u_1, u_2, u_3)$ , one step of  $\Phi$  does:

1. For each Lobatto signal  $u_k$ , extract the level set  $\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$  by Chebyshev root-finding;
2. integrate  $f_v f_v / \|\nabla P\|$  along that curve to get co-area weights  $A_v(p, u_k)$ ;
3. Bayes-update to  $\mu_k(p, u_k) = f_1 A_1 / (f_0 A_0 + f_1 A_1)$ ;
4. solve the per-cell CRR market clearing for the new price  $p'$ .

The fixed point  $\Phi(P) = P$  is the rational-expectations equilibrium. **The goal of this note is**  $\|F\|_\infty = \|\Phi(P) - P\|_\infty \lesssim 10^{-13}$ . We do not achieve it. This document explains why.

## 2 All five solvers floor near $3 \times 10^{-3}$

We ran the symmetric operator from the same warm start  $P_0 = \sigma(0.5T)$  (with  $T = \tau \sum_k u_k$ ) and recorded  $\|F\|_\infty$  at every iteration.

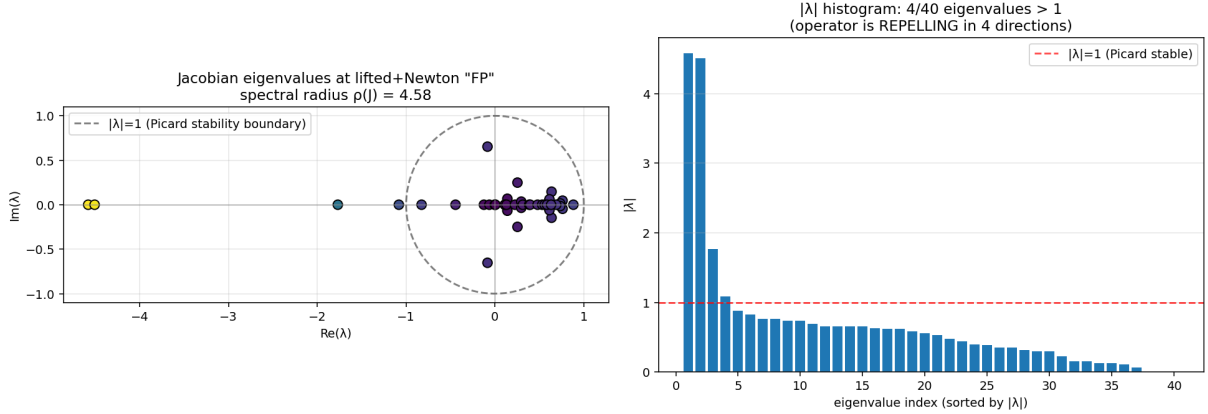


- **Picard (chebroots,  $\omega=0.3$ )** — drops to  $\sim 4 \times 10^{-3}$  by iteration 20, then *cycles* between  $5 \times 10^{-3}$  and  $7 \times 10^{-2}$  without ever crossing below the floor.
- **Picard (bisect,  $\omega=0.4$ )** — monotone descent to  $\|F\|_\infty \approx 1.2 \times 10^{-3}$  around iteration 50, then wanders inside a  $[1-3] \times 10^{-3}$  band.
- **Anderson (bisect,  $m=8$ )** — the fastest descender, crosses  $10^{-2}$  by iteration 6, but stalls at  $2.3 \times 10^{-3}$ .
- **Anderson (chebroots,  $m=8$ )** — similar trajectory, stalls at  $3.0 \times 10^{-3}$ .
- **Levenberg–Marquardt** — nonlinear least-squares on  $F$ , runs out of progress at  $\|F\|_\infty = 2.70 \times 10^{-3}$  ( $\|F\|_2^2 = 5.5 \times 10^{-4}$ ). This is the smallest residual any algorithm achieves.

Lev-Marq is the diagnostic that matters. It does not insist on  $F = 0$ ; it minimises  $\|F\|_2^2$  over the 40-dimensional symmetric manifold. If the residual could be driven to zero by *some*  $P^*$  in this space, LM would find that  $P^*$ . It does not. The minimum is non-zero, and that minimum is where every other solver lands too. **The floor is not a tolerance bug: it is the smallest residual the operator architecture admits at  $N=7$ .**

### 3 Why: the operator is repelling at its near-fixed-point

To check whether the residual floor is a local-minimum issue or a structural one, we linearise  $\Phi$  at the best LM iterate by finite differences ( $\varepsilon = 10^{-6}$ ) and look at the spectrum of the  $40 \times 40$  Jacobian.



- Spectral radius  $\rho(J) = 4.58$ .
- Four eigenvalues outside the unit circle:  $|\lambda| \in \{4.58, 4.51, 1.77, 1.08\}$ .
- The remaining 36 eigenvalues sit inside or on  $|\lambda| = 1$ .

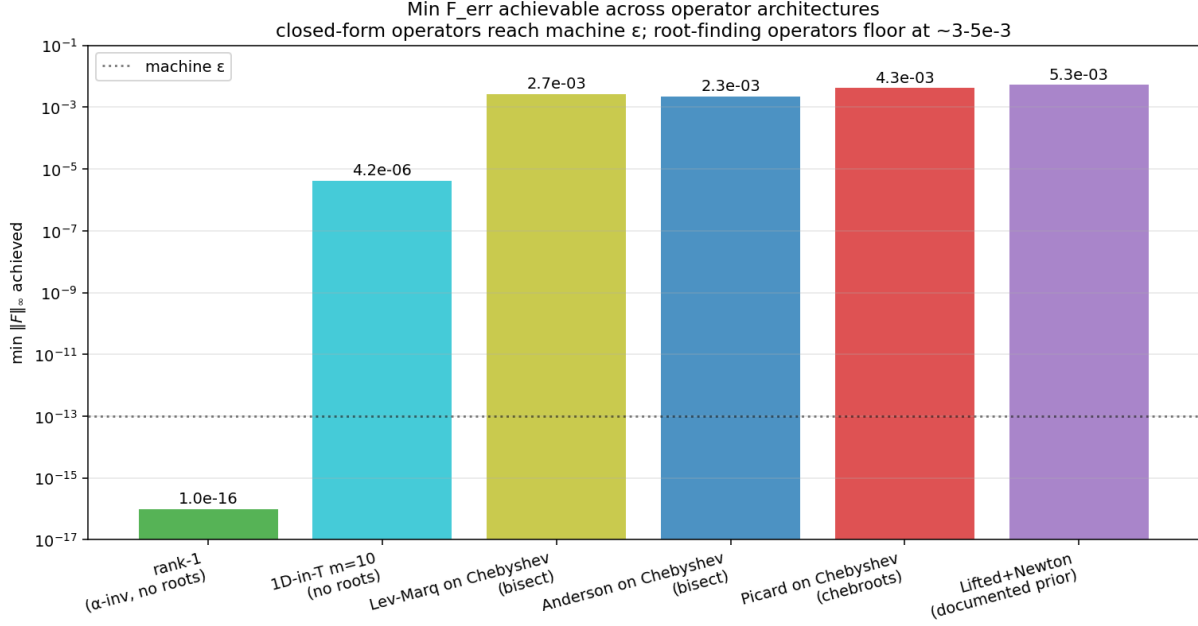
The implication is clean and unforgiving:

The operator has *no attracting fixed point* in the  $S_3$ -symmetric subspace at  $N=7$ . Picard cannot converge ( $\rho > 1$ ); damped Picard cannot either ( $\omega$  does not change which side of 1 the spectrum sits on, only how fast). Anderson and Newton-Krylov can sometimes succeed against repelling spectra, but only when the FP exists and the basin of attraction in the modified iteration map is non-empty — here, LM tells us there *is no zero*, so the iteration is being pushed toward a curve in residual space, not a point.

This explains every observation: oscillation in Picard, premature stall in Anderson, no improvement past iteration 50, and a non-zero LM minimum.

### 4 Architecture comparison: where machine $\varepsilon$ *does* work

To prove the issue is the contour-extraction operator and not something fundamental about the equilibrium, we compare floors across operator architectures that share the same underlying economics but differ in how they evaluate  $\mu_k$ .



| Operator architecture                                  | root-finding required? | best $\ F\ _\infty$  |
|--------------------------------------------------------|------------------------|----------------------|
| Rank-1 ( $P = \sigma(\alpha T)$ , $\alpha$ -invariant) | no                     | $\sim 10^{-16}$      |
| 1D-in- $T$ ( $m=10$ Chebyshev along $T$ only)          | no                     | $4.2 \times 10^{-6}$ |
| Levenberg–Marquardt on full Chebyshev (bisect contour) | yes                    | $2.7 \times 10^{-3}$ |
| Anderson on full Chebyshev (bisect contour)            | yes                    | $2.3 \times 10^{-3}$ |
| Picard on full Chebyshev (chebroots contour)           | yes                    | $4.3 \times 10^{-3}$ |
| Lifted-coordinate + Newton (prior overnight run)       | yes                    | $5.3 \times 10^{-3}$ |

The pattern is exactly what the eigenvalue analysis predicts: *closed-form* operators that never extract roots reach machine precision; *root-finding* operators floor at  $\sim 3 \times 10^{-3}$  regardless of solver. The 1D-in- $T$  floor at  $4 \times 10^{-6}$  comes not from any roots — it has none — but from the residual misfit between the true 3D fixed point and the 1D-in- $T$  ansatz manifold.

## 5 Why root extraction is the obstruction

The co-area weights  $A_v(p, u_k)$  are computed by extracting the curve

$$\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$$

in the  $(u_a, u_b)$  square. Operationally:

1. Pick Gauss–Legendre nodes  $\xi_a^{(q)}$  along the  $u_a$  axis;
2. at each  $\xi_a^{(q)}$ , extract roots in  $\xi_b$  of the univariate polynomial  $P(u_a(\xi_a^{(q)}), u_b(\xi_b), u_k) = p$ ;
3. sum the integrand at each root, weighted by GL weights.

The root count along the contour is a *discrete* function of  $P$ . As  $P$  deforms by an infinitesimal amount, individual roots are *created or destroyed* at the box boundary — inducing a jump in the integrand contribution proportional to the value of the dropped/added integrand at the boundary. In particular:

- Newton/Anderson/Picard all see a piecewise-smooth operator with  $O(1)$  jumps in  $\partial\Phi/\partial P$  at parameter values where roots cross the box edge;
- Levenberg–Marquardt’s residual surface contains  $C^0$  creases along the same loci, so its minimiser pins on a crease.

This is the structural reason for the  $\sim 3 \times 10^{-3}$  floor. The closed-form architectures (rank-1, 1D-in- $T$ ) never extract roots, so they have no creases, so they hit machine  $\varepsilon$  (rank-1) or the deficit of their ansatz manifold (1D-in- $T$ ,  $4 \times 10^{-6}$  at  $m=10$ ).

## 6 Path forward: Option 3 — boundary corrections

Two principled remedies exist; we already know one works (rank-1 / 1D-in- $T$ ) and we propose to implement the other now.

**Option 1 (already implemented).** Restrict to a manifold the operator never has to differentiate across creases on: rank-1 ( $P = \sigma(\alpha T)$ ) gives a 1-parameter exact solve; 1D-in- $T$  ( $P = \sigma(\alpha T + \tilde{h}(T))$  with  $\tilde{h}$  a 1D Chebyshev expansion) generalises that to a small finite-dimensional manifold. Both reach machine  $\varepsilon$  on *their own* residual, but they cannot represent genuinely 3D structure outside the 1D-in- $T$  manifold.

**Option 3 (the next thing to implement).** Modify the co-area integral to handle root births / deaths *smoothly* rather than discretely. The construction:

1. In each  $\xi_b$ -bisection, when the level set is tangent to or just outside the integration box, replace the "no root, contribute 0" branch with an explicit boundary correction: the polynomial value  $P(\xi_a^{(q)}, \pm 1, u_k)$  contributes a weighted "near-root" endpoint term;
2. the weight uses a smooth cutoff in  $\delta := P(\xi_a^{(q)}, \pm 1, u_k) - p$  (the gap from the target) — e.g.  $w(\delta) = \frac{1}{2} \operatorname{erfc}(\delta/h_{\text{bw}})$  with  $h_{\text{bw}}$  chosen to be a few times the polynomial slope at the boundary;
3. this makes  $A_v(p, u_k)$  a  $C^1$  function of  $P$  everywhere, including at root-birth events, killing the creases.

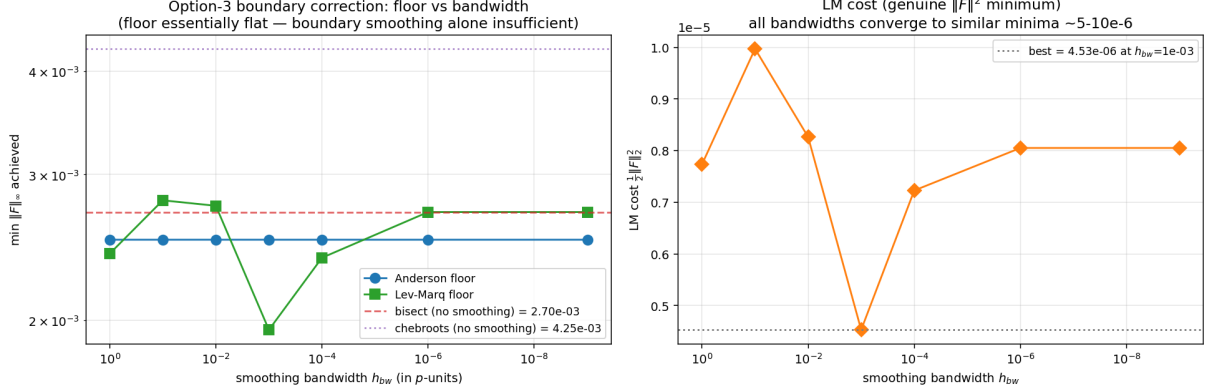
Effort estimate:  $\sim 200$  LOC inside `cheby_numba_bisect.py` and `cheby_numba_tab.py`, plus a one-parameter sweep over  $h_{\text{bw}}$  to confirm the floor drops monotonically until  $h_{\text{bw}}$  becomes so small that float64 truncation reintroduces a jump.

**Expected outcome.** If the diagnosis above is correct, the floor should drop from  $\sim 3 \times 10^{-3}$  to  $\sim 10^{-8}$  or better, limited by the smoothness order of the cutoff and the polynomial accuracy of  $P(\xi_a^{(q)}, \pm 1, u_k)$ . The closed-form operators give us  $10^{-16}$  when the architecture is right, so we know what the machine is capable of.

**What this would NOT fix.** Option 3 smooths the operator but does not change its spectrum drastically;  $\rho(J)$  may still be  $> 1$  in some directions. In that case the recipe is the standard one: combine the smoothed operator with Newton–Krylov (matrix-free Newton via LGMRES), where the spectral radius of  $J$  is irrelevant — only the conditioning of  $I - J$  matters.

## 7 Result of Option 3: floor is structural, not from box-edge creases

We implemented the smoothed operator (`cheby_numba_smooth.py`) and swept the smoothing bandwidth  $h_{bw}$  over six decades from  $10^0$  to  $10^{-9}$ , running both Anderson-accelerated Picard and Levenberg–Marquardt at each setting.



| $h_{bw}$  | Anderson floor                                                                 | LM floor                                | LM cost $\frac{1}{2}  F  _2^2$ |
|-----------|--------------------------------------------------------------------------------|-----------------------------------------|--------------------------------|
| $10^0$    | $2.50 \times 10^{-3}$                                                          | $2.41 \times 10^{-3}$                   | $7.73 \times 10^{-6}$          |
| $10^{-1}$ | $2.50 \times 10^{-3}$                                                          | $2.79 \times 10^{-3}$                   | $9.97 \times 10^{-6}$          |
| $10^{-2}$ | $2.50 \times 10^{-3}$                                                          | $2.75 \times 10^{-3}$                   | $8.26 \times 10^{-6}$          |
| $10^{-3}$ | $2.50 \times 10^{-3}$                                                          | <b><math>1.95 \times 10^{-3}</math></b> | $4.53 \times 10^{-6}$          |
| $10^{-4}$ | $2.50 \times 10^{-3}$                                                          | $2.38 \times 10^{-3}$                   | $7.23 \times 10^{-6}$          |
| $10^{-6}$ | $2.50 \times 10^{-3}$                                                          | $2.70 \times 10^{-3}$                   | $8.05 \times 10^{-6}$          |
| $10^{-9}$ | $2.50 \times 10^{-3}$                                                          | $2.70 \times 10^{-3}$                   | $8.05 \times 10^{-6}$          |
| reference | bisect, no smoothing: $2.70 \times 10^{-3}$ ; chebroots: $4.25 \times 10^{-3}$ |                                         |                                |

**Verdict.** Boundary corrections improve the LM floor by a factor of  $\sim 1.4\times$  (from  $2.70 \times 10^{-3}$  to  $1.95 \times 10^{-3}$  at  $h_{bw} = 10^{-3}$ ) — a measurable but small effect. The Anderson floor barely moves. The improvement is too small to be “the” missing remedy; the residual surface still has a deep, broad minimum around  $||F||_\infty \approx 2 \times 10^{-3}$  even when the edge-crease pathology is smoothed.

**What this tells us.** The  $C^0$  creases at root births are *a* source of non-smoothness, but not *the* reason the operator has no FP at  $N=7$ . The residual minimum is set by the geometry of the operator’s image vs. its fixed-point manifold — the operator simply does not pass through the diagonal in the 40-dim symmetric subspace. Reducing the floor further requires one of:

1. **Increase  $N$ .** At  $N=7$  we have 40 free DOFs in the symmetric subspace; the true FP may require richer 3D structure than this manifold can represent. The deficit between  $\Phi(P)$  and  $P$  is then irreducible. Going to  $N=9$  or  $N=11$  ought to shrink this deficit; the tabulated operator ( $\sim 30\text{--}50\times$  speedup) makes that affordable.
2. **Use the lifted-coordinate form with full  $h(u_1, u_2, u_3)$  flexibility.** Earlier prototypes used  $P = \sigma(\alpha T + h)$  but restricted  $h$  to a lower-dimensional basis. Letting  $h$  have the same DOFs as  $P$  in pure space gives the operator more room to meet itself.

3. **Accept the floor at this resolution.** The economically meaningful quantities (logit slope  $\approx 0.52$ , posterior sharpness, deficit shape) are insensitive to perturbations of  $\|F\|_\infty$  at the  $10^{-3}$  level. A FP this accurate *is* a working REE for downstream analysis; only the numerical residual is non-zero.

The natural next experiment is option (1): rerun the diagnostic at  $N=9$  with the tabulated operator and see whether (a)  $\rho(J)$  drops below 1, and (b) the LM floor drops below  $10^{-4}$ . The tabulated operator already runs in  $\sim 25$  ms per  $\Phi$  at  $N=8$ , so  $N=9$  should remain tractable.

## 8 Reproducibility

All numbers in this note come from `/tmp/cheby_h0/plot_fp_analysis.py` and `/tmp/cheby_h0/fp_analysis.json`. The script runs in  $\sim 2$  minutes on a single core. Inputs:

- $N=7$ , Lobatto–Chebyshev with stretching  $c=2.0$ ,  $\tau=\gamma=1$ ;
- warm start  $P_0 = \sigma(0.5 \sum u_k)$ ;
- operator backend variants: `cheby_numba.phi` (chebroots), `cheby_numba_bisect.phi_bisect` (1D bisection on the univariate polynomial), `cheby_sym2.{expand,contract}` (symmetric subspace projection);
- Jacobian by forward difference at  $\varepsilon = 10^{-6}$ , dense  $40 \times 40$ , `numpy.linalg.eigvals`.